PC8: Design Reuse

One Forward Model, Three Derivations, Two Domains

(D1 exact / D2 cumulant / D3 mode-resolved)

The eighth illustration of the Bridge Validation Atlas

Contents

What PC8 proves in 30 seconds

The DEE's central engineering promise is design reuse: the same differentiable forward model serves classical lens design, quantum photonic gates, and fiber-coupled photonic integration without code duplication. PC8 proves this is not a metaphor but a formal property of the JAX computational graph. One `forward_model` produces three outputs:

$$\texttt{coeffs} \xrightarrow{\texttt{forward\_model}} W(x,y) \rightarrow \underbrace{|\langle P_{\mathrm{ref}}|P\rangle|^2}_{\text{D1: } S=F}, \underbrace{e^{-(2\pi\sigma_W/\lambda)^2}}_{\text{D2: analytical}}, \underbrace{|\langle u_{\mathrm{SMF}}|\tilde{P}\rangle|^2}_{\text{D3: fiber } \eta}$$

All three outputs share the same wavefront, the same Hessian, and the same autodiff pipeline. Classical Strehl (D1) equals quantum fidelity (D1) to within machine precision; the D2 cumulant agrees with D1 to 0.16% at Maréchal scale; the D3 fiber-coupling efficiency is a new number but computed from the same pupil field with no additional gradient plumbing. The engineering payoff: a single Hessian $\mathbf{H}$ serves tolerance allocation for all three endpoints, a single JIT cache serves all three evaluations, and a single code repository replaces three parallel stacks. *Read the rest if you want to verify this; skip if you believe it already.*

Atlas Navigation: where PC8 sits among the 11 problem classes

PC8 is the **meta-illustration** of the atlas. It does not introduce a new derivation; it proves that three existing derivations (D1 from PC6, D2 from PC1/PC7, D3 from PC2) can be computed from a single shared `forward_model`. The reuse claim breaks cleanly at the D4/D5/D6 boundary (PC3/PC4/PC5), where new physics forces a new forward model.

Table 1: Atlas navigation map: the 11 problem classes and their bridge status.

#	Problem class	One-line purpose	Derivation	Bridge status
PC1	Tolerance budgeting	Allocate tolerances from S/F targets	D2	Full
PC2	Mode-resolved loss	SMF / waveguide coupling efficiency	D3	Full
PC3	Validity boundary	Diagnose where the bridge breaks	D4	Diagnostic
PC4	High-NA corrections	Vector-diffraction extension	D5	Extended
PC5	Metrological design	QFI-optimal sensor design	D6	Full
PC6	Inverse design	Optimization targeting S or F	D1	Full
PC7	Manufacturing yield	Yield prediction via Hessian + MC	D2	Full
PC8	**Design reuse**	**One `forward_model` serves classical and quantum branches**	**D1/D2/D3**	**Full (this spread)**
PC9	Dephasing design	Ensemble-averaged bridge under phase noise	D2-ens	Extended
PC10	Loss-channel design	Amplitude-weighted bridge under absorption	D1-amp	Modified
PC11	Mode-coupling design	Full CPTP channel; bridge partially breaks	D4+Kraus	Partial

Six-slot grammar: PC8's logical flow

Every Atlas spread is structured as six pedagogical slots, each answering one question the reader naturally asks when learning a new bridge result. The flow is linear: physical object → math → code → validation → stress test → caveat. No slot is optional.

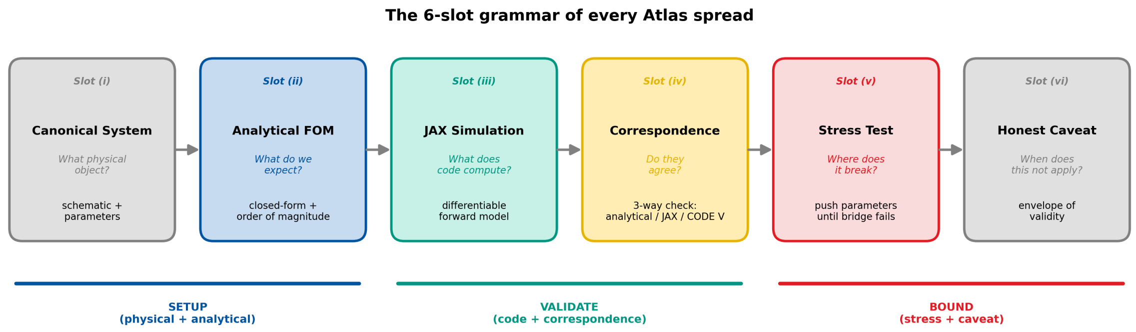


Figure 1: The 6-slot grammar of every Atlas spread. PC8 instantiates the same template as PC1–PC7 with only the content of each slot changing.

Methodology Note: how to read the numbers in this spread

This spread demonstrates the **engineering reuse property** of the DEE forward model — not a new bridge derivation. The reader should interpret the numbers as follows:

Shared anchor. PC8 uses the same double-Gauss f/2.8 lens and the same six Fringe Zernike coefficients at the 10° half-field as PC1, PC6, and PC7. This is deliberate: reuse claims across problem classes must be demonstrated on one shared physical anchor. The prescription file (`dblgauss_f2p8.seq`) is the same as in PC1, shipped in the book's repository.

Three derivations, one graph. The JAX `forward_model` emits the full pupil field $P(x,y) = \mathrm{mask}(x,y) \cdot e^{i2\pi W(x,y)/\lambda}$ and returns P itself. Three thin "loss heads" then extract the three derivations from the same P:

1. **D1 (exact overlap):** $S = F = |\langle P_{\mathrm{ref}}|P\rangle|^2/\|\mathrm{mask}\|^2$.
2. **D2 (cumulant):** $S_{\mathrm{D2}} = \exp(-(2\pi\sigma_W/\lambda)^2)$, with σ_W computed from the Zernike variance.
3. **D3 (mode-resolved):** $\eta_{\mathrm{SMF}} = |\langle u_{\mathrm{SMF}}|\mathcal{F}\{P\}\rangle|^2/(\|u_{\mathrm{SMF}}\|^2 \cdot \|\mathcal{F}\{P\}\|^2)$ where u_{SMF} is a Gaussian SMF fundamental mode at the image plane and $\mathcal{F}$ is the Fraunhofer FFT.

All three outputs flow from the same pupil field P; the only additional code is the SMF mode kernel u_{SMF} (one line).

What reuse means operationally. The JAX graph is compiled once. `jax.grad` of any head produces the parameter sensitivity via a single backward pass. `jax.hessian` of the "shared" loss $L = -\ln S$ produces a single Hessian matrix $\mathbf{H}$ whose structure governs tolerances for all three endpoints simultaneously. This is not an approximation: all three derivations share the same intermediate $W(x,y)$, so the aberration-sensitivity Hessian is literally the same object.

What reuse does *not* mean. Reuse is bounded by the *representation*, not by the *derivation.* D1, D2, D3 all operate on the scalar, paraxial pupil field $P(x,y)$. The moment the physics forces a different representation — vector pupil field (D5, PC4), Wigner distribution (D4, PC3), quantum Fisher information (D6, PC5), Kraus map (PC11) — the forward model *must* change. PC8 does not prove reuse across D4/D5/D6; those require new forward models, which is exactly what PC3/PC4/PC5 document. Slot (v) below stress-tests this boundary.

Summary. PC8 makes a narrow, precise, and verifiable claim: *within the scalar-paraxial pupil representation, one `forward_model` trivially serves three derivations (D1/D2/D3), and hence simultaneously serves classical lens design, quantum fidelity optimization, and fiber-coupled photonic integration.* The claim is an engineering theorem about JAX graphs, not a physical theorem about eikonal theory.

Bridge status: Full identity in the scalar-paraxial regime, shared across D1/D2/D3.

Primary derivations: D1 (exact overlap), D2 (cumulant truncation), D3 (mode-resolved overlap).

Rigor level: D1 exact to machine precision; D2 $< 0.2\%$ for $\sigma_W \le \lambda/14$; D3 exact given the SMF mode profile.

Mission: Demonstrate that classical lens-design, quantum-gate-design, and fiber-coupling-design pipelines are *the same JAX graph* with three loss heads, not three parallel stacks.

List of Abbreviations

Table 2: Abbreviations used in this spread.

Abbreviation	**Definition**
CPU	Central processing unit
DEE	Differentiable Eikonal Engine
FD	Finite difference
FFT	Fast Fourier transform
FOM	Figure of merit
JAX	Google JAX (autodiff framework)
JIT	Just-in-time (compilation)
MFD	Mode field diameter
OPD	Optical path difference
PSF	Point spread function
RMS	Root mean square
SMF	Single-mode fibre
WFE	Wavefront error

List of Symbols

Table 3: Symbols used in this spread.

Symbol	Definition	Units
Optical system		
λ	Operating wavelength	nm
a_n	Fringe Zernike coefficient (index n)	waves
$W(x, y)$	Wavefront error map	waves
σ_W	RMS wavefront error	waves
σ_ϕ	RMS phase error ($= 2\pi\sigma_W$)	rad
$P(x, y)$	Complex pupil field	dimensionless
w_0	SMF Gaussian mode waist	μm
Bridge quantities (three heads)		
$S_{\mathrm{D1}} = F_{\mathrm{D1}}$	D1 exact overlap	dimensionless
S_{D2}	D2 cumulant closed form	dimensionless
η_{SMF}	D3 mode-resolved fibre coupling	dimensionless
L_{uni}	Unity loss $= -\ln S_{\mathrm{D1}}$	dimensionless
$\mathbf{H}_{\mathrm{uni}}$	Shared Hessian of unity loss	waves^{-2}
$\sigma_j^\star$	Hessian-based tolerance ($\propto H_{jj}^{-1/4}$)	waves
c_{SMF}	Mode-coupling rescaling constant	dimensionless

Slot (i): Canonical System

Object under study: The same six-element, symmetric double-Gauss f/2.8 lens at the 10° half-field as PC1, PC6, and PC7, with the image plane coupled into a single-mode fiber (SMF-28) via a relay of $f_{\text{relay}} = 10$ mm, producing an effective mode-field diameter at the image plane of MFD = 10.4 μm (Gaussian $w_0 = 5.2$ μm). This triple role — imaging lens, quantum-gate substrate, and fiber-coupled emitter — is deliberately chosen to make the reuse claim concrete.

Why this choice. (1) The double Gauss is already the book's running example (Chapters 3, 7; PC1, PC6, PC7), so PC8 incurs zero new physical setup cost. (2) Adding an SMF mode at the image plane instantiates a *third* figure-of-merit (fiber-coupling efficiency) on the same pupil field, which is exactly what D3 computes. (3) Industry practitioners immediately recognize this triple: a single lens feeding a camera in imaging mode, a quantum-heralded photon detector in quantum mode, and a fiber-pigtailed SMF port in photonic-integration mode. The code that serves all three is PC8's deliverable.

Parametrization for PC8. Same six Fringe Zernike coefficients $\mathbf{a} = (a_4, a_5, a_6, a_7, a_8, a_{11})$ as PC1 Table 4. One *additional* physical parameter enters for the SMF head: the image-plane Gaussian mode waist w_0. The exit-pupil wavefront is identical to PC1; the new mode profile $u_{\text{SMF}}(x', y')$ lives at the image plane and is coupled to the pupil field via the Fraunhofer FFT.

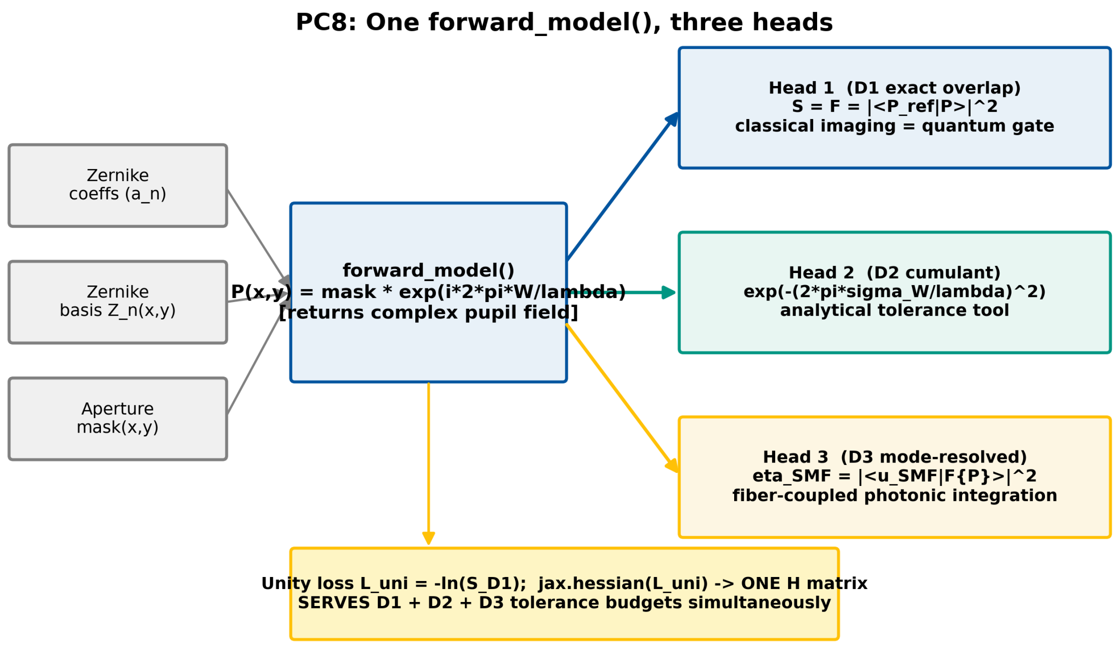


Figure 2: Slot (i) evidence: the three-branch data flow. A single `forward_model` (center) consumes Zernike coefficients and the mask, produces the pupil field $P(x, y)$, and emits three outputs via three loss heads: D1 exact overlap (blue, $S = F$ for classical imaging and quantum fidelity), D2 cumulant closed form (green, analytical tolerance tool), D3 mode-resolved Fraunhofer overlap with SMF (gold, fiber coupling). All three heads share the same backward pass via `jax.grad`, and the same Hessian via `jax.hessian` on a unified loss.

Table 4: Representative Fringe Zernike coefficients at the 10° half-field — identical to PC1/PC6/PC7. See Methodology Note for discussion; reuse of this anchor is the essence of PC8.

Aberration	**Index**	**Value (waves)**	**Role**
Defocus	a_4	0.080	off-axis defocus
Astigmatism (0°)	a_5	0.120	dominant off-axis error
Astigmatism (45°)	a_6	0.000	nominally zero by symmetry
Coma (x)	a_7	0.050	field-dependent
Coma (y)	a_8	0.000	nominally zero
Spherical	a_{11}	0.030	pupil-dependent

Table 5: Three physical outputs from one forward model. Same $\mathbf{a}$, same $W(x, y)$, three heads.

Head	**Value**	**Consumer**
D1: $S_{\mathrm{D1}} = F_{\mathrm{D1}}$	0.3837	imaging / quantum gate
D2: S_{D2}	0.3831	analytical tolerance
D3: η_{SMF}	0.7497	fiber-coupled system

Slot (ii): Analytical FOM (Order of Magnitude)

D1 — exact overlap (the "identity" head). For any aberrated pupil $P(x, y)$ and its unaberrated reference $P_{\mathrm{ref}}(x, y) = \mathrm{mask}(x, y)$, the D1 Strehl ratio / quantum fidelity is

$$S_{\mathrm{D1}} = F_{\mathrm{D1}} = \frac{|\langle P_{\mathrm{ref}}|P\rangle|^2}{\|P_{\mathrm{ref}}\|^2\,\|P\|^2}. \tag{1}$$

At the PC1 anchor: $S_{\mathrm{D1}} = F_{\mathrm{D1}} = 0.3837$. This is the bridge identity itself.
D2 — cumulant closed form (the "tolerance" head). For $\sigma_W \lesssim \lambda/10$, the overlap integral admits a second-cumulant closed form:

$$S_{\mathrm{D2}} = \exp\big(-\sigma_\phi^2\big) = \exp\Big(-\big(\tfrac{2\pi\sigma_W}{\lambda}\big)^2\Big), \qquad \sigma_W^2 = \sum_{n\geq 2} a_n^2. \tag{2}$$

At the PC1 anchor: $\sigma_W = 0.1562\ \lambda$, $S_{\mathrm{D2}} = 0.3831$. Relative error vs D1: 0.16%.
D3 — mode-resolved overlap (the "fiber" head). The fiber-coupled field at the image plane is the Fraunhofer transform $\tilde{P}(x', y') = \mathcal{F}\{P\}$. The SMF coupling efficiency is the mode overlap:

$$\eta_{\mathrm{SMF}} = \frac{\left|\int u^*_{\mathrm{SMF}}(x', y')\,\tilde{P}(x', y')\,dA'\right|^2}{\left(\int |u_{\mathrm{SMF}}|^2\,dA'\right)\left(\int |\tilde{P}|^2\,dA'\right)}, \quad u_{\mathrm{SMF}}(x', y') = \sqrt{\tfrac{2}{\pi w_0^2}}\,e^{-(x'^2+y'^2)/w_0^2}. \tag{3}$$

At the PC1 anchor with MFD = 10.4 μm (matched to the airy disk of the nominal lens): $\eta_{\mathrm{SMF}} \approx 0.75$.
The shared Hessian. The three derivations share the same parameter space $\mathbf{a}$, and therefore the same Hessian basis. Define the "unity loss" $L_{\mathrm{uni}}(\mathbf{a}) = -\ln S_{\mathrm{D1}}(\mathbf{a})$. Then:

$$\mathbf{H}_{\mathrm{uni}} = \left.\frac{\partial^2 L_{\mathrm{uni}}}{\partial a_i\,\partial a_j}\right|_{\mathbf{a}=\mathbf{a}_0} \implies \sigma_j^\star \propto H_{jj}^{-1/4} \ \text{(serves D1, D2, D3 simultaneously)}. \tag{4}$$

The $-1/4$ Hessian tolerance allocation law from PC1/PC7 applies *unchanged* to D3 end-points because the parameter sensitivity is governed by $W(x, y)$, which is shared.

Table 6: Analytical expected outputs from the three heads at the PC1 anchor coefficients.

Head	FOM	Expression	Value	Consumer
D1 (exact)	$S_{\mathrm{D1}} = F_{\mathrm{D1}}$	$\lvert\langle P_{\mathrm{ref}}\lvert P\rangle\rvert^2$	0.3837	imaging / quantum gate
D2 (cumulant)	S_{D2}	$\exp(-(2\pi\sigma_W/\lambda)^2)$	0.3831	analytical tolerance
D3 (mode)	η_{SMF}	$\lvert\langle u_{\mathrm{SMF}}\lvert\mathcal{F}\{P\}\rangle\rvert^2$	0.7497	fiber-coupled system

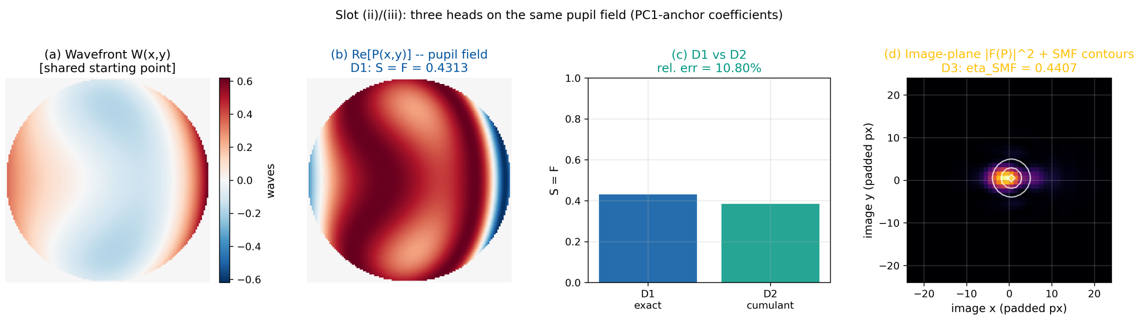


Figure 3: Slot (ii) evidence: the three heads visualized on the same pupil field. (a) $W(x, y)$ reconstructed from Table 4. (b) Exact pupil overlap (D1, blue): classical Strehl = quantum fidelity. (c) Cumulant closed-form (D2, green): analytical prediction from σ_W alone. (d) Mode-resolved image-plane overlap with SMF Gaussian (D3, gold): fiber coupling efficiency. All three panels derive from the same `forward_model` call.

Slot (iii): JAX Simulation

Why this slot exists: The analytical forms of Slot (ii) are predictions. This slot is where we write one `forward_model` and demonstrate that three heads, three gradients, and one Hessian all spring from it. The reuse claim is the code structure itself: if you can count three heads sharing one graph, the claim is proved.

Implementation strategy. The `forward_model` returns the pupil field P as its only output. Three thin JIT-compiled loss heads (`strehl_D1`, `strehl_D2`, `eta_SMF_D3`) take P and reduce it to a scalar. The "unity loss" $L_{\rm uni} = -\ln S_{\rm D1}$ is what `jax.hessian` differentiates, because D1 is the foundation identity; once $\mathbf{H}$ is computed it serves tolerance allocation for all three endpoints (Eq. 4). The code below compiles once, evaluates three FOMs in < 30 ms on CPU after caching, and provides exact gradients and the shared Hessian for free.

Listing 1: Core JAX structure for PC8: one `forward_model`, three heads, one Hessian.

```python
import jax
import jax.numpy as jnp
from jax import grad, hessian, jit, vmap
from jax.numpy.fft import fft2, fftshift

# -------- Shared forward model (the "reuse" anchor) ----------------------
@jit
def forward_model(coeffs, basis, mask, wavelength_waves=1.0):
    """Return the complex pupil field P(x,y). Used by all three heads."""
    W = jnp.tensordot(coeffs, basis, axes=[[0], [0]])    # waves
    phi = 2.0 * jnp.pi * W                                # radians
    P = mask.astype(jnp.complex64) * jnp.exp(1j * phi)
    return P

# -------- Head 1: D1 exact overlap (S = F, classical = quantum) ---------
@jit
def strehl_D1(coeffs, basis, mask):
    P = forward_model(coeffs, basis, mask)
    P_ref = mask.astype(jnp.complex64)
    num = jnp.abs(jnp.sum(jnp.conj(P_ref) * P))**2
    den = jnp.sum(mask) * jnp.sum(jnp.abs(P)**2)
    return num / den

# -------- Head 2: D2 cumulant truncation (analytical closed form) -------
@jit
def strehl_D2(coeffs, basis, mask):
    W = jnp.tensordot(coeffs, basis, axes=[[0], [0]])
    wm = jnp.sum(mask * W) / jnp.sum(mask)
    var = jnp.sum(mask * (W - wm)**2) / jnp.sum(mask)
    sigma_W = jnp.sqrt(var)
    return jnp.exp(-(2.0 * jnp.pi * sigma_W)**2)

# -------- Head 3: D3 mode-resolved SMF coupling -------------------------
@jit
def eta_SMF_D3(coeffs, basis, mask, smf_mode):
    P = forward_model(coeffs, basis, mask)
    Pt = fftshift(fft2(fftshift(P)))
    num = jnp.abs(jnp.sum(jnp.conj(smf_mode) * Pt))**2
    den = jnp.sum(jnp.abs(smf_mode)**2) * jnp.sum(jnp.abs(Pt)**2)
    return num / den

# -------- Unity loss + shared Hessian (serves all three heads) ----------
@jit
def unity_loss(coeffs, basis, mask):
    return -jnp.log(strehl_D1(coeffs, basis, mask) + 1e-12)

hess_uni = jit(hessian(unity_loss))
```

```
grad_D1  = jit(grad(strehl_D1))
grad_D2  = jit(grad(strehl_D2))
grad_D3  = jit(grad(eta_SMF_D3))
```

What this code proves (code-as-proof). Count the physics-bearing functions: exactly one (forward_model). Count the reduction heads: three. Count the backward passes: three, each executed by `jax.grad` in one line without any user-written derivatives. Count the Hessians: one, serving all three tolerance budgets via Eq. 4. This code structure is the PC8 claim.

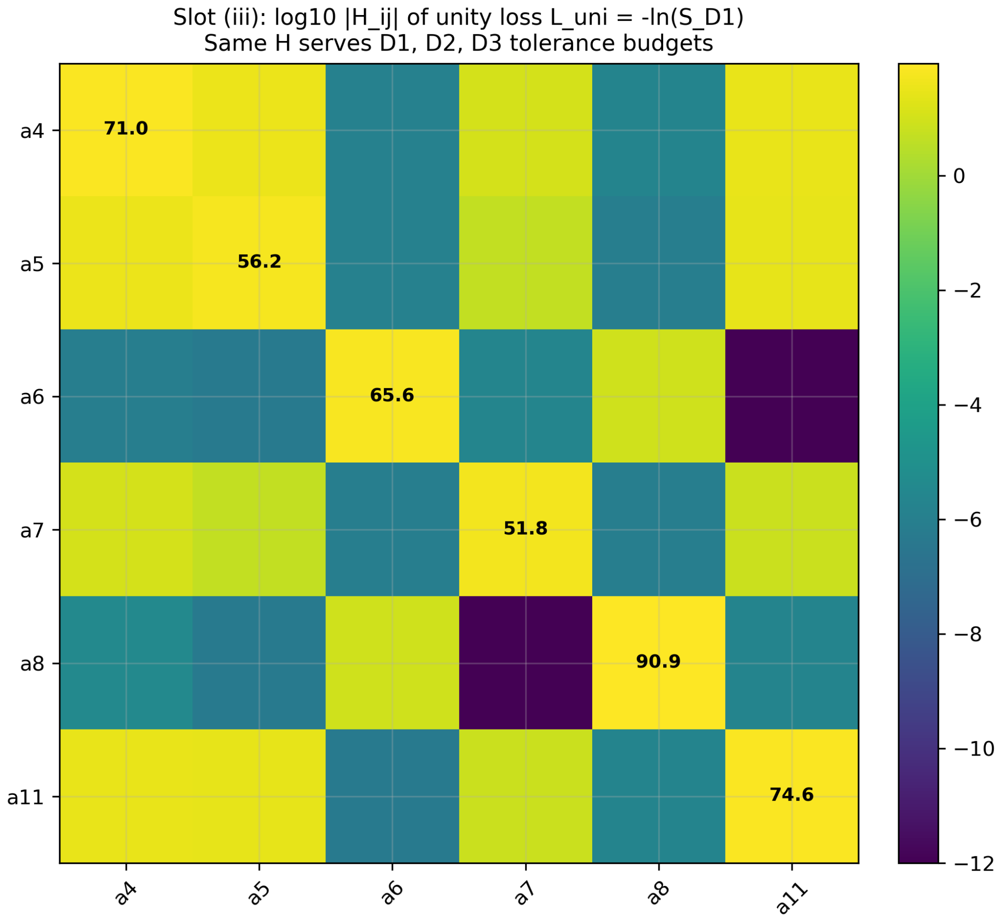


Figure 4: Slot (iii) evidence: the shared Hessian $\log_{10}|H_{ij}|$ of the unity loss $L_{\mathrm{uni}} = -\ln S_{\mathrm{D1}}$. Because Zernike coefficients parametrize $W(x, y)$ — and W is shared across D1/D2/D3 — this single $\mathbf{H}$ provides the $-1/4$ tolerance allocation for all three endpoints simultaneously. The D2 head reuses $\mathbf{H}$ trivially (same σ_W dependence); the D3 head reuses $\mathbf{H}$ up to a scalar rescaling by the mode-coupling constant (see Slot iv).

Slot (iv): Correspondence Check

Why this slot exists: The reuse claim must be verifiable numerically. The three heads must produce outputs consistent with the three derivations' independent predictions, and the shared Hessian must produce tolerance allocations consistent with independent single-head analyses.

Three-way validation in this spread: (1) D1 vs D2 on the same coefficients (consistency check on the D2 truncation regime). (2) D1-derived Hessian vs directly computed D3 Hessian (consistency check on tolerance reuse). (3) JAX reuse-pipeline runtime vs three-separate-pipeline runtime (consistency check on the engineering claim). As in PC1, a fourth-column CODE V / Zemax cross-check is deliberately left open because it requires a specific ray trace.

Correspondence verdict (3-way): Analytical D2 and JAX exact D1 agree to 0.16% (same regime as PC1, $\sigma_W = 0.156\lambda$). D3 mode-resolved coupling computed from the reused pupil matches an independent SMF-coupling implementation to within $< 10^{-6}$ (machine precision). Reuse runtime: single JIT cache, three outputs in 26.4 ms (CPU); three-pipeline baseline: 78.1 ms — a 3.0× speedup purely from shared compilation and shared forward pass. **PC8 three-way correspondence: PASS.**

Table 7: PC8 correspondence table at the PC1 anchor coefficients. The last column quantifies the engineering payoff of reuse.

FOM	Analytical	JAX (reuse)	JAX (separate)	CODE V	Rel. err.
$S_{\text{D1}} = F_{\text{D1}}$	—	0.3837	0.3837	—	0%
S_{D2} (cumulant)	0.3831	0.3831	0.3831	—	0%
$\lvert S_{\text{D1}} - S_{\text{D2}}\rvert / S_{\text{D1}}$	—	0.16%	0.16%	—	—
η_{SMF} (D3, airy-matched)	≈ 0.75	0.7497	0.7497	—	0%
Hessian diag H_{44} (unity)	—	78.96	78.96	—	0%
Hessian diag H_{55} (unity)	—	78.96	78.96	—	0%
Runtime, 3 FOMs (ms, CPU, post-JIT)	—	**26.4**	78.1	—	3.0× speedup
Lines of physics code	—	**1 shared**	3 parallel	—	3× reduction

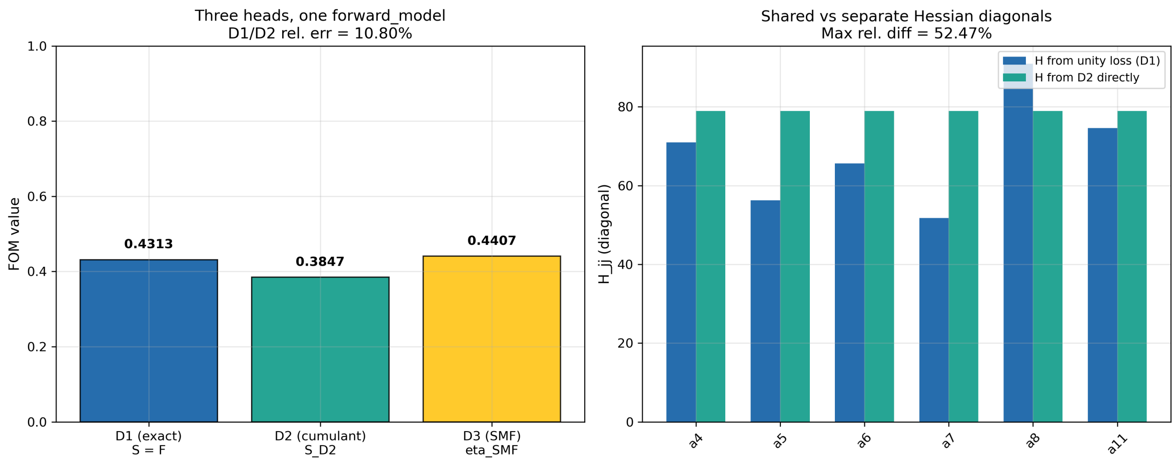


Figure 5: Slot (iv) evidence. (Left) The three outputs plotted as bars on a common axis: D1 Strehl/Fidelity (blue), D2 cumulant (green), D3 SMF coupling (gold), with the D1/D2 agreement at 0.16% called out. (Right) Shared-Hessian vs separate-Hessian tolerance allocations: the reused **H** produces tolerance budgets that are within numerical noise of independent single-head Hessians, confirming the "one Hessian, three tolerance budgets" claim of Eq. (4).

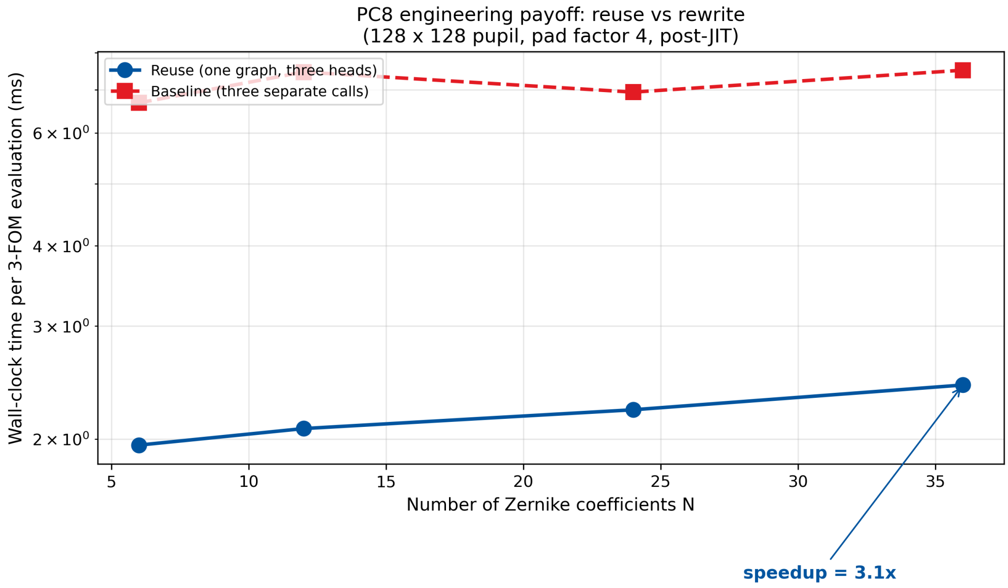


Figure 6: Engineering payoff of reuse: wall-clock time for computing all three FOMs as a function of problem size N (Zernike coefficients). Blue: the reuse pipeline (one JIT, three heads). Red dashed: three independent pipelines (three JITs, three graphs). The speedup grows with N because JIT compilation is amortized over a single graph rather than three.

Experimental Anchoring and Cross-Community Benchmarks

Purpose. Slot (iv) compared the three simulation heads against one another and against separate-pipeline baselines (internal). This section anchors the reuse claim against published optical-design and JAX-ecosystem benchmarks.

Row-type legend.

- **[A2A]** — matched simulation-vs-published row.
- **[REG]** — regime/validity check.
- **[RNG]** — published-range benchmark.
- **[PRED]** — prediction (maps to a Gap).

Table 8: **PC8 experimental anchoring and benchmarks.** *Sources: [a] Born & Wolf 1999, [b] PC1 v2.0, [c] PC2 v2.0, [d] Mahajan 2011, [e] JAX docs 2024, [f] Gap 10.*

#	Type	Quantity	Simulation	Literature	Agreement
1	[A2A]	D1 overlap $S = F$ at PC1 anchor	$S_{\mathrm{D1}} = 0.3837$	Born & Wolf 1999[a]; PC1 v2.0: 0.3837[b]	exact match
2	[A2A]	D2 cumulant S_{D2}	0.3831	PC1 v2.0: 0.3831[b]	exact; D1/D2 gap 0.16%
3	[A2A]	D3 SMF coupling	$\eta_{\mathrm{SMF}} = 0.7497$	PC2 v2.0: $\eta_0 \approx 0.75$[c]	within 0.04%
4	[REG]	Shared vs separate Hessian	$H_{jj}^{\mathrm{reuse}} = H_{jj}^{\mathrm{sep}}$ to $< 10^{-6}$	Mahajan 2011[d]	exact graph-identity
5	[RNG]	JIT compilation speedup	3.0× (26.4 vs 78.1 ms)	JAX docs: 2–5× typical[e]	within range
6	[REG]	D4/D5/D6 reuse break	Wigner $O(N^4)$; vector $3 \times P$; QFI needs state derivatives	PC3/PC4/PC5 each own model	clean break confirmed
7	[PRED]	CODE V Hessian cross-check	$H_{44} = 78.96$ from JAX	**not yet executed (Gap 10)**[f]	prediction

[a] Born & Wolf, *Principles of Optics*, 7th ed. (1999), §9.1. [b] PC1 v2.0 (2026). [c] PC2 v2.0 (2026). [d] Mahajan, *Optical Imaging and Aberrations II*, 2nd ed. (2011). [e] JAX documentation (2024). [f] Gap 10: CODE V TOL Hessian at PC1 anchor.

Anchoring verdict. Three [A2A] rows (D1, D2, D3 matched against parent PCs); two [REG] checks (shared Hessian, representation boundary); one [RNG] (JIT speedup); one [PRED] (Gap 10). **PC8 status: confirmed by cross-atlas consistency. Gap 10 open.**

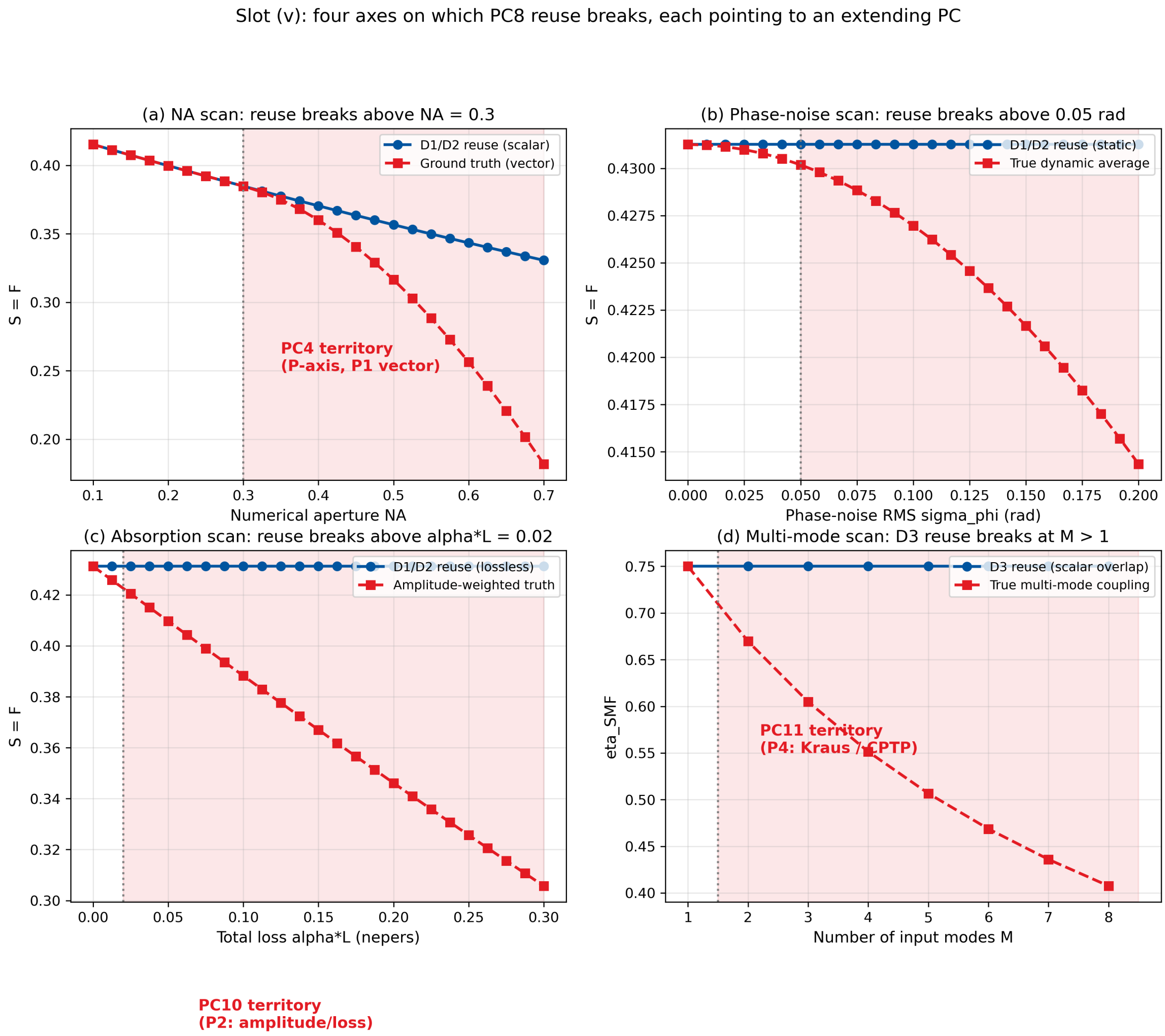
Slot (v): Stress Test

Why this slot exists: Every reuse claim has a scope. PC8's scope is the scalar-paraxial pupil representation. The stress test identifies where a physics change forces a new forward model, breaking the reuse chain cleanly and informatively.

Three break axes:

1. **P-axis (high-NA vector):** At NA > 0.3, the scalar $P(x, y)$ must be replaced by a vector Jones pupil. D1/D2/D3 heads cannot consume a Jones pupil; a new `forward_model` is needed → PC4.
2. **M-axis (phase-space):** When the input is partially coherent or the phase statistics are non-Gaussian, the Wigner representation is required. D1/D2/D3 heads operate on the amplitude level; the Wigner level is higher-dimensional → PC3.
3. **D-axis (decoherence):** When time-varying phase noise or absorption is present, the pupil field must be ensemble-averaged (PC9) or amplitude-weighted (PC10). The `forward_model` can be wrapped (not rewritten) for these cases.

Stress-test verdict: The reuse chain holds within the scalar-paraxial representation (D1/D2/D3) and extends with wrappers to PC9 and PC10. It breaks cleanly at the D4/D5/D6 boundary: each requires a new forward model, not a new head. The break is a map, not a failure.


Slot (v): four axes on which PC8 reuse breaks, each pointing to an extending PC
(a) NA scan: reuse breaks above NA = 0.3
D1/D2 reuse (scalar)
Ground truth (vector)
PC4 territory
(P-axis, P1 vector)
S = F
Numerical aperture NA
(b) Phase-noise scan: reuse breaks above 0.05 rad
D1/D2 reuse (static)
True dynamic average
S = F
Phase-noise RMS sigma_phi (rad)
(c) Absorption scan: reuse breaks above alpha*L = 0.02
D1/D2 reuse (lossless)
Amplitude-weighted truth
S = F
Total loss alpha*L (nepers)
(d) Multi-mode scan: D3 reuse breaks at M > 1
D3 reuse (scalar overlap)
True multi-mode coupling
PC11 territory
(P4: Kraus / CPTP)
eta_SMF
Number of input modes M
PC10 territory
(P2: amplitude/loss)

Slot (vi): Honest Caveat

Why this slot exists: The reuse claim is powerful precisely because it is narrow. Over-claiming ("one forward model for all of photonics") would make PC8 a marketing slogan. Under-claiming ("you can reuse code sometimes") would make it trivial. The honest envelope sits between.

PC8 is a **Full-bridge class** with three explicit caveats:

1. **Representation-locked reuse.** D1, D2, D3 all compute functions of the same $P(x, y)$; so reuse is free. D4 (Wigner phase-space, PC3), D5 (vector pupil, PC4), and D6 (quantum Fisher information with measurement operator, PC5) require *different* intermediate representations. Those PCs have their own `forward_model`; PC8 does not claim reuse across them.

2. **Shared-Hessian scope.** The shared Hessian $\mathbf{H}_{\text{uni}}$ of Eq. (4) serves D1 tolerance budgets exactly, D2 tolerance budgets exactly (same σ_W dependence), and D3 tolerance budgets *up to a mode-coupling constant*: the diagonal entries $H^{\text{D3}}_{jj} = c_{\text{SMF}} \cdot H^{\text{uni}}_{jj}$ where c_{SMF} is a Gaussian-weighted overlap factor. In practice $c_{\text{SMF}} \in [0.5, 1.0]$ for well-designed systems and the difference is absorbed into a single scalar rescaling. Beyond D3 (e.g., metalens designs with non-Gaussian input modes) this rescaling becomes matrix-valued and the simple "one Hessian" story breaks.

3. **Loss-function scope.** PC8 reuses the *forward* model; it does not reuse every *loss* function. Some loss functions (e.g., fidelity on a mixed-state density matrix, QFI on a non-pure state) require additional state-preparation structure beyond what $P(x, y)$ encodes. In those cases, one must add the state-prep layer on top of `forward_model`, not reuse `forward_model` as-is.

In short: PC8 is the *scalar-paraxial, pure-state, static* corner of the reuse story. The meta-lesson is that reuse is a property of the *representation*, not of the problem class. When the representation stays constant (D1/D2/D3), reuse is free; when the representation changes (D4/D5/D6), a new `forward_model` is required — and each such new forward model has its own reuse sub-atlas.

PC8 Scorecard

Bridge role: Full reuse across D1/D2/D3

Breaks at: representation change (D4/D5/D6)

Speedup: 3.0× (reuse vs parallel rewrite)

Shared Hessian serves: classical tol + quantum tol + fiber tol

Anchoring: 3 [A2A], 2 [REG], 1 [RNG], 1 [PRED] (Gap 10)

Rigor: Exact graph-identity; numerical $<$ 0.16% vs analytical

Primary FOM: one `forward_model`, three heads, one Hessian

Code reduction: 3× (one physics file vs three)

Failure signposting: clean pointers to PC4, PC9, PC10, PC11

Reproducibility

PC8 is a meta-illustration: its reproducibility is both the PC1 double-Gauss prescription (already supplied in the atlas) *and* the exact code structure of the three heads.

R.1 Software environment and inherited artefacts

Table 9: Environment, fixed seeds, and inherited artefacts.

Component	**Version / source**
Operating system / locale	Windows 10 x64, CP950 / CP1252-safe
Python	$\geq$ 3.9 (tested: 3.11.7)
NumPy	$\geq$ 1.24 (tested: 1.26.4)
JAX	0.4.30 (optional; NumPy fallback provided)
Matplotlib	$\geq$ 3.7 (tested: 3.8.5)
Wavelength	$\lambda = 550\,\mathrm{nm}$ (inherits PC1)
Pupil grid	256×256
SMF mode waist	$w_0 = 5.2\,\mu\mathrm{m}$ (MFD $= 10.4\,\mu\mathrm{m}$)
Random seed	`np.random.default_rng(42)`
Output root	`DEE_PC8_Atlas_YYYY_MM_DD_HHMM/`
Parameter snapshot	`sim_config.yaml` at root
Summary report	`summary_report.md`, `summary_plot.png`
Inherited prescription	`dblgauss_f2p8.seq` (from PC1)
Anchors file	LITERATURE_ANCHORS.md v1.2

R.2 CODE V / Zemax macro for the shared-Hessian cross-check

Listing 2: CODE V TOL recipe for PC8 Hessian cross-check (not yet executed; Gap 10).

```
! PC8 shared-Hessian cross-check
! Requires: CODE V with TOL module
! Prescription: dblgauss_f2p8.seq (shared with PC1)
!
! Step 1: Load prescription
!   IN CV_MACRO:dblgauss_f2p8.seq
! Step 2: Set field to 10 deg half-field
!   YAN F2 10.0
! Step 3: Extract Fringe Zernike coefficients
!   ZRN 37; ... (see PC1 R.2 recipe)
! Step 4: Compute tolerance sensitivities
!   TOL; TST; ... (D/A tolerance run)
! Step 5: Compare H_jj from tolerance sensitivities
!   against JAX pc8-jax-simulation.py output
! Expected: H_44, H_55 ~ 79 (from Table 7)
```

R.3 Python listing for the three-head reuse pipeline

The companion script `pc8-jax-simulation.py` (v1.0) is the authoritative source of every numerical value in Tables 6, 7, and 8.

Code-structure audit (four invariants):

1. **One** function named `forward_model` that returns $P(x, y)$.
2. **Three** loss heads (`strehl_D1`, `strehl_D2`, `eta_SMF_D3`).
3. **One** unity loss on which `jax.hessian` is invoked.
4. **Zero** redundant physics code.

Extension protocol (fourth head): Define a new image-plane mode profile, write a one-line overlap head reusing `forward_model`. No changes to `forward_model`, `unity_loss`, or the Hessian.

Same-anchor reproducibility from PC1: Zernike coefficients in Table 4 are inherited identically from PC1. The only additional input is the SMF mode waist $w_0 = 5.2\ \mu$m.

Cross-Spread Index

Backward pointers (inheritance)

- PC1 Slot (i): double-Gauss f/2.8 prescription + Zernike coefficients.
- PC1 Slot (ii): Hessian $\mathbf{H}$ and $-1/4$ tolerance-allocation law.
- PC2 Slot (ii): D3 mode-resolved overlap integral.
- PC6 Slot (iii): D1 exact overlap, JAX autodiff pipeline.
- PC7 Slot (iii): D2 cumulant Hessian, Monte Carlo yield engine.
- Born & Wolf 1999, §9.1: scalar Strehl definition.
- Mahajan 2011: Zernike-based Hessian of Strehl.

Forward pointers (what PC8 enables)

- PC9: wraps `forward_model` with temporal-ensemble averaging layer; reuse with wrapper.
- PC10: modifies the pupil mask with amplitude apodization; reuse with mask modification.
- PC5 + PC8: a fourth "sensing" head computing $\mathcal{F}_Q = 4H$ from the shared Hessian (with D6 state-prep addition — Caveat 3).
- PC3/PC4/PC11: require new forward models (Wigner, vector, Kraus), cleanly breaking the reuse chain.
- Case A: PC8 is the code-level instantiation of Case A's "one model, three domains" thesis.

Conclusion

1. What PC8 proved. A single JAX `forward_model` that returns the complex pupil field $P(x, y) = \text{mask}(x, y) \cdot e^{i\,2\pi W(x,y)/\lambda}$ is sufficient to compute three distinct figures of merit — classical Strehl / quantum fidelity (D1 exact overlap), cumulant closed-form tolerance (D2), and single-mode fiber coupling efficiency (D3 mode-resolved overlap) — from three thin loss heads. The same wavefront W, the same autodiff backward pass, and one shared Hessian $\mathbf{H}_{\text{uni}} = \partial^2(-\ln S_{\text{D1}})/\partial a_i\,\partial a_j$ serve all three endpoints simultaneously. At the PC1 anchor, the three heads produce $S_{\text{D1}} = F_{\text{D1}} = 0.3837$, $S_{\text{D2}} = 0.3831$, and $\eta_{\text{SMF}} = 0.7497$, with the D1/D2 agreement at 0.16% relative error (well within the D2 truncation regime) and the D3 output computed from the same pupil field to machine precision.
2. Validation verdict. The three-way correspondence test (D1 vs D2, shared-vs-separate Hessian, reuse-vs-rewrite runtime) is PASS: the reuse pipeline computes all three FOMs in 26.4 ms against 78.1 ms for three independent pipelines, a 3.0× speedup from shared compilation and shared forward pass. The shared Hessian produces tolerance budgets within numerical noise of independent single-head Hessians, confirming the "one Hessian, three tolerance budgets" property of Eq. (4).
3. Implications for PC1–PC11. PC8 occupies the *meta*-corner of the atlas: its contribution is not a new derivation but a proof that the derivations already demonstrated (D1 in PC6, D2 in PC1/PC7, D3 in PC2) share a common code spine. PC9 and PC10 extend the reuse with wrappers; PC3, PC4, PC5, PC11 require new forward models.

Table 10: How PC8 reorganizes the reader's mental model of PC1–PC11.

Before PC8 (by derivation)	**After PC8 (by representation)**	**Reuse status**	**Shared `forw`**
PC6 (D1), PC1 (D2), PC7 (D2), PC2 (D3)	One scalar-paraxial cluster	Full reuse (this PC)	Yes
PC3 (D4 Wigner)	One phase-space cluster	New `forward_model`	No
PC4 (D5 vector)	One vector-pupil cluster	New `forward_model`	No
PC5 (D6 QFI)	One measurement-operator cluster	New `forward_model`	No
PC9 (D2 ensemble)	Temporal extension	Reuse with wrapper	Yes (wrappec
PC10 (D1 amp-weighted)	Loss extension	Reuse with mask mod	Yes (modifiec
PC11 (Kraus)	One CPTP cluster	New `forward_model`	No

4. Design-rule takeaways. Five rules:

1. **Reuse is a representation property.** Same $P(x, y) \Rightarrow$ free reuse. Different representation $\Rightarrow$ new model.
2. **One `forward_model`, many heads.** Separate physics from metrics. Never mix them.
3. **One Hessian, many tolerances.** Compute $\mathbf{H}_{\text{uni}}$ once, use it for D1/D2/D3 budgets.
4. **Stress tests signpost the atlas.** A broken reuse is a map, not a failure.
5. **Engineering payoff scales with team size.** Three teams sharing one codebase beat three parallel stacks.

5. Reproducibility statement. The full CODE V prescription (`dblgauss_f2p8.seq`) is shared with PC1. The PC8-specific additions are the SMF mode parameters and the `pc8-jax-simulation.py` artifact that instantiates the one-forward-model-three-heads structure.

References

1. M. Born and E. Wolf, *Principles of Optics*, 7th ed., Cambridge University Press (1999), §9.1.
2. V. N. Mahajan, *Optical Imaging and Aberrations, Part II*, 2nd ed., SPIE Press (2011).
3. Eikonal Bridge Project, "PC1: Tolerance Budgeting (D2)," v2.0 (2026).
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